

Triage and priority-based healthcare diagnosis using artificial intelligence for autism spectrum disorder and gene contribution: A systematic review

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ABSTRACT

The exact nature, harmful effects and aetiology of autism spectrum disorder (ASD) have caused widespread confusion. Artificial intelligence (AI) science helps solve challenging diagnostic problems in the medical field through extensive experiments. Disease severity is closely related to triage decisions and prioritisation contexts in medicine because both have been widely used to diagnose various diseases via AI, machine learning and automated decision-making techniques. Recently, taking advantage of high-performance AI algorithms has achieved accessible success in diagnosing and predicting risks from clinical and biological data. In contrast, less progress has been made with ASD because of obscure reasons. According to academic literature, ASD diagnosis works from a specific perspective, and much of the confusion arises from the fact that how AI techniques are currently integrated with the diagnosis of ASD concerning the triage and priority strategies and gene contributions. To this end, this study sought to describe a systematic review of the literature to assess the respective AI methods using the available datasets, highlight the tools and strategies used for diagnosing ASD and investigate how AI trends contribute in distinguishing triage and priority for ASD and gene contributions. Accordingly, this study checked the Science Direct, IEEE Xplore Digital Library, Web of Science (WoS), PubMed, and Scopus databases. A set of 363 articles from 2017 to 2022 is collected to reveal a clear picture and a better understanding of all the academic literature through a final set of 18 articles. The retrieved articles were filtered according to the defined inclusion and exclusion criteria and classified into three categories. The first category includes 'Triage patients based on diagnosis methods' which accounts for 16.66% ($n = 3/18$). The second category includes 'Prioritisation for Risky Genes' which accounts for 66.6% ($n = 12/18$) and is classified into two subcategories: 'Mutations observation based', 'Biomarkers and toxic chemical observations'. The third category includes 'E-triage using telehealth' which accounts for 16.66% ($n = 3/18$). This multidisciplinary systematic review revealed the taxonomy, motivations, recommendations and challenges of ASD research that need synergistic attention. Thus, this systematic review performs a comprehensive science mapping analysis and discusses the open issues that help perform and improve the recommended solution of ASD research direction. In addition, this study critically reviews the literature and attempts to address the current research gaps in knowledge and highlights weaknesses that require further research. Finally, a new developed methodology has been suggested as future work for triaging and prioritising ASD patients according to their severity levels by using decision-making techniques.

1. Introduction

Autism spectrum disorder (ASD) is a neuropsychiatric disorder that affects brain activities at different levels with high evidence of genetic contribution. According to the World Health Organization (WHO), 67 million persons are affected with ASD around the world [1]. Autistic children manifest many symptoms, such as repetitive behaviour,

abnormalities in facial and voice expressions and slow speech learning, as well as communication difficulties [2]. These symptoms generally are observed during the first two years and tend to be continuous. The aetiology of this kind of disorder are varied and unclear, some cases are related to gene mutations, such as "de novo or transmitted loss of function (LOF) mutations" [3], or it could be caused by environmental toxic involvement in the genetic abnormalities [4]. Some types of autism have been inherited from one or both parents. Males could be affected

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List of abbreviations*Acronyms*
Definition

ADOS	Autism diagnostic observation schedule	IT2TR-FWZIC	Interval type-2 trapezoidal-fuzzy weighted with zero inconsistency combined
ADI-R	Autism diagnostic interview-revised	lncRNAs	Long non-coding RNAs
AHP	Analytical hierarchy process	LOF	Loss of function
AI	Artificial intelligence	LR	Logistic regression
ANP	Analytical network process	MCDM	Multi-criteria decision making
ASD	Autism spectrum disorder	MEW	Multiplicative exponential weighting
BWM	Best–worst method	ML	Machine learning
CNNs	Convolutional neural networks	MRI	Magnetic resonance imaging
CT	Computed tomography	NA	Is reveals that the study doesn't specify any of used matrix
DL	Deep learning	PANDA	Prioritisation of autism-genes using network-based deep-learning approach
DSM-IV-TR	Diagnostic and Statistical Manual of Mental Disorders Text Revision Fourth Edition	PTAP	Processes for triaging autism patient
EEG	Electroencephalogram	RF	Random forest
FDOSM	Fuzzy decision by opinion score method	RITA-T	Rapid interactive screening test for autism in toddlers
FDM	Fuzzy Delphi method	SAW	Simple additive weighting
FRN	Brain-specific functional relationship network	SFARI	Simons Foundation Autism Research Initiative
FWZIC	Fuzzy-weighted zero-inconsistency	SVM	Support vector machines
GBA	Guilt by association	TADA	Transmission and De Novo Association Analysis
HAW	Hierarchical adaptive weighting	TFN	Triangular fuzzy number
ICNs	Is intrinsic connectivity networks	WHO	World Health Organization
		WPM	Weighted product model
		WSM	Weighted sum model

with these neurodevelopmental disorders four times more than females [5]. As such, autism must be diagnosed at early stages to help limit the development of the condition [6]. Therefore, many clinicians depend on traditional diagnostic methods, such as Autism Diagnostic Observation Schedule (ADOS), to observe the autistic patient behaviour and Autism Diagnostic Interview-Revised (ADI-R), which takes a long time to complete diagnosis and make decisions. Several computer-aided auxiliary diagnostic devices, such as electroencephalogram (EEG), magnetic resonance imaging (MRI) and computed tomography (CT), have been utilised for this purpose. The application of CT is not recommended because it exposes the patient to potentially harmful ionizing radiation. Therefore, the use of this imaging technology requires careful risk assessment. In addition, the high cost and time consumption for other devices, such as MRI, where a child often needs to be quiet or in a general anaesthesia to allow accurate imaging, carry inherent risks [5]. Many researchers believed that genes and mutations are highly associated with ASD, and they still attempted to discover what types and priority of gene sets exactly affect autism to obtain an accurate diagnosis. This confusion motivated us to shed light on the different trends and perspectives and attempted to paint an appropriate picture of how to integrate the concepts of triage of ASD, as well as the prioritisation of genes that contribute towards the diagnosing of ASD by using AI techniques. Accordingly, the introduction covers three important questions according to the scope of our study and provides appropriate answers.

On the one hand, an early and timely diagnosis is highly associated with the triage concept that is based on patients' prioritisation. Thus, the first question that needs to be determined is: **'What is triage and prioritisation and why is it important for autistic patients?'**

In medical use, the degrees of urgency of triage is assigned to wounds or illnesses to decide the order of treatment for a large number of sick or injured people [7]. Triage is a patient's urgent situation level while the prioritisation is ranking patients according to their severity levels [8]. The two terms are still confusing and interrelated to each other. However, the concept of triage is the process used to identify the most urgent cases and to determine who should be treated and in what order, whereas prioritising is (required) to rank or list a set of things (i.e. patients, genes, features) in order of priority or importance [9,10]. For example, triage could be assigned to patients with four severity levels, such as risk, urgent, sick and normal whereas prioritisation can be

applied within each severity level to rank patients according to their health situation [11]. Thus, for the first procedure, relying on triage in clinics is important to determine the ASD patient with the highest urgency needs. For the second procedure, patients can have priority for clinical appointment and treatment to cut waiting times and make the best use of available resources. The two terms differ slightly in meaning but are quite different in medical applications; therefore, this confusion needs to be explained, especially in autistic research.

Triage levels can be assigned for ASD patients who cover different levels, such as severe, moderate and mild cases.

[5]. In addition, patients can be grouped on the basis of a triage using a triage-based questionnaire to identify appropriate ASD patients who need priority diagnostic evaluation [9]. Although some autism centres, such as the Marcus Autism Centre and UMN-AND, opted to prioritise patients according to some criteria (e.g. young children, or kids with more urgent needs or give a priority to those with additional complexities) that can impact treatment decisions [9]. Besides, the concept of prioritising in gene contexts has been widely used to prioritise the list of genes that can be a cause of ASD, such as prioritising genes to calculate their importance in a "cell-type-specific" [12], which can be etiological to ASD. In conclusion, triage and priority tools support for leading autism diagnosis, understanding evidence-based diagnosis, which is used in ASD medical tests to guide treatment decisions and provide more explanations to make the autism medical contexts and its treatment more accurate [10].

On the other hand, many researchers attempted to leverage AI development technology in ASD diagnosis cases; therefore, the second question is **'How can AI contribute to ASD diagnosis?'**

AI is placed on triage and priority for machines to learn from medical experience, adjust to new inputs and perform physician-like tasks [13]. Most medical AI applications that are presented today – from heart chronic disease, diabetics, Alzheimer's, etc. – rely heavily on data mining, machine learning, natural language processing, intelligent and decision making, more. Using these technologies, computers can be trained to accomplish triage and prioritisation strategies and/or tasks for different ASD directions (e.g., genes priority, patients' triage, patients' priority) by processing large amounts of autistic data and recognizing diagnosis patterns in the data. AI software can be programmed to accurately spot signs of a certain ASD in medical images, such as MRIs

and CT scans [14]. Existing similar solutions have already been studied by researchers to use AI for reducing time and obtaining a concise ASD diagnosis. By using such AI tools, clinicians can diagnose patients more precisely and prescribe the most suitable healthcare. Amongst others, machine learning has been widely used to determine the accuracy metrics utilised in diagnosis. According to the literature, several ML methods, such as support vector machines (SVM), random forest (RF), and others had been used in ASD diagnosis to distinguish the risk ASD and non-ASD genes [15,16]. Trends shown in this question are restricted to some AI directions, sparse and scattered throughout the ASD literature. Accordingly, this paper attempts to update the literature on ASD diagnosis of ambiguity systematically using AI techniques and associated concepts of triage and priority. Many of the published reviews have been addressed ASD but they are not complementary to AI, triage, priority and diagnosis in telehealth and/or telemedicine. Therefore, the third question is **‘What are the current academic reviews of literature of the ASD triage and prioritisation based AI, and what is critical analysis?’**

To determine the current state of academic knowledge about prioritising ASD, the aim of previous studies and limitations should be discussed. In Ref. [17], the authors summarised 13 studies where AI is used to assess and examine whether other behavioural data can potentially be used to recognize ASD characteristics. Although the machine learning algorithms depend on the training dataset, no extensive research assesses how the quality of input data can affect the accuracy yet. While the ethical issue around data privacy is also another problem under debate, reports on AI technology integrated with autism disorder issues are not efficient. The survey of [18] discussed how individuals with ASD were handled by technology-based intervention methodologies, including modern computer-aided systems, computer vision-assisted technologies and virtual reality. They concluded that a gap exists between neuroscience, including autism disorders, psychology and the research community of computer science to develop an AI technology for ASD individuals. The 31 collected studies conclude that clinicians were not convinced in general with technology-based interventions because of the non-empirical nature of results that had been published. The major reason is a greater number of studies on distinct intervention methodologies does not obey any specific standard or AI research design. In Ref. [19], the study aimed to identify and describe supervised ML directions in ASD articles and guide researchers in expanding their research computationally and statistically towards this disease. The limitations they found related to unlabeled data were consuming time, cost and effort besides the complicity of some ML models. Thus, the researchers must consider if the understanding of models is going to be more important than the result when selecting the ML model. In conclusion to the above reviews, the authors neglected the diagnosis towards the importance of triage and prioritisation in autism patients’ life. At this point, to fill up the comparative analyses between the presented work and published reviews, Table 1 shows the comparative points and differences.

As shown in Table 1, most previous works did not cover the main aspects that the present study aims to provide. Some reviews covered some important aspects but neglected other concepts. In addition, different important information in this systematic review is explained represented by tables, figures, discussion and analysis considering

integration amongst health diagnosis autism, triage and priority in ASD, gene contributions, the AI technologies for supporting ASD, and telehealth/telemedicine application for helping ASD. No such study has been presented yet. In this aspect, this study has multifaceted contributions as follows: (1) Constructing new taxonomy to classify the available studies for the triage and prioritisation of ASD into three distinct categories. (2) Highlights the motivations, challenges, recommendations and open issues and key solutions that need focus to reduce the complexity of health autism diagnosis through AI technologies. (3) Explain various dataset results and available sources to embrace triage and priority for ASD and present useful tables that contain all datasets and machine learning methods found in the literature. (4) Offering the innovation of our ideas that developed a multi-criteria decision making (MCDM)-based used several methods (e.g., Delphi method, IT2TR-FWZIC method, and FDOSM) that can provide for triage autism patients or even prioritise them according to their severity levels.

2. Method

This study was formed according to the systematic review guidelines showed in (Fig. 1) [20,21] and collected the literatures based on query from multiple reliable databases: Science direct (SD), IEEE Xplore digital library (IEEE), Web of Science (WoS), PubMed and Scopus databases. The collected articles focused on ASD triage and prioritisation and gene contributions from different perspectives aided by AI.

2.1. Search strategy

This study utilised a Boolean query (conclude OR, AND) to link the keywords (‘autism’ OR ‘autistic’) AND (‘triage’ OR ‘prioritisation’ OR ‘prioritisation’ OR ‘telemedicine’) AND (‘artificial intelligence’ OR ‘multi-criteria decision making’ OR ‘machine learning’ OR ‘deep learning’). The study applied the query to ensure collecting of all related papers based on our topic besides promoting the study with other papers that focused on how the triage affected the ASD patients. The collected articles from databases were published from 2017 to 2022.

2.2. Inclusion criteria and exclusion criteria

All written papers in the English language and submitted to conferences or journals were included. The main topics in the studies are related to ASD that utilised various topics, such as prioritisation, ranking and triage with machine learning algorithms and AI techniques. Papers with the telemedicine and telehealth perspectives are also included. The exclusion criteria consisted of all papers that were not written in the English language. The diagnosis of autism without a clear method, tool, strategy and approach for prioritising and triaging cases were also excluded.

3. Comprehensive science mapping analysis

The application of bibliometric measurements in various fields has been increased. Several techniques and software tools have been proposed to implement the scientific mapping analysis of published studies of fragmented and controversial streams of research [22–25]. The total

Table 1
Comparisons between existing related reviews and the present systematic research.

Ref.	Type of paper	Year coverage	Databases used	Taxonomy based	PRISMA statement	Integrated triage/priority, and genes diagnosis using AI
[17]	Literature Review	2009–2019	PubMed	✓	✓	✓
[18]	Survey	Since 1973	N/A	✓	✓	✓
[19]	Review	2005–2018	Endnote and Google Scholar	✓	✓	✓
The presented study	Systematic Review	2017–2022	Web of Science, IEEE Explore, Scopus, Science Direct, PubMed	✓	✓	✓

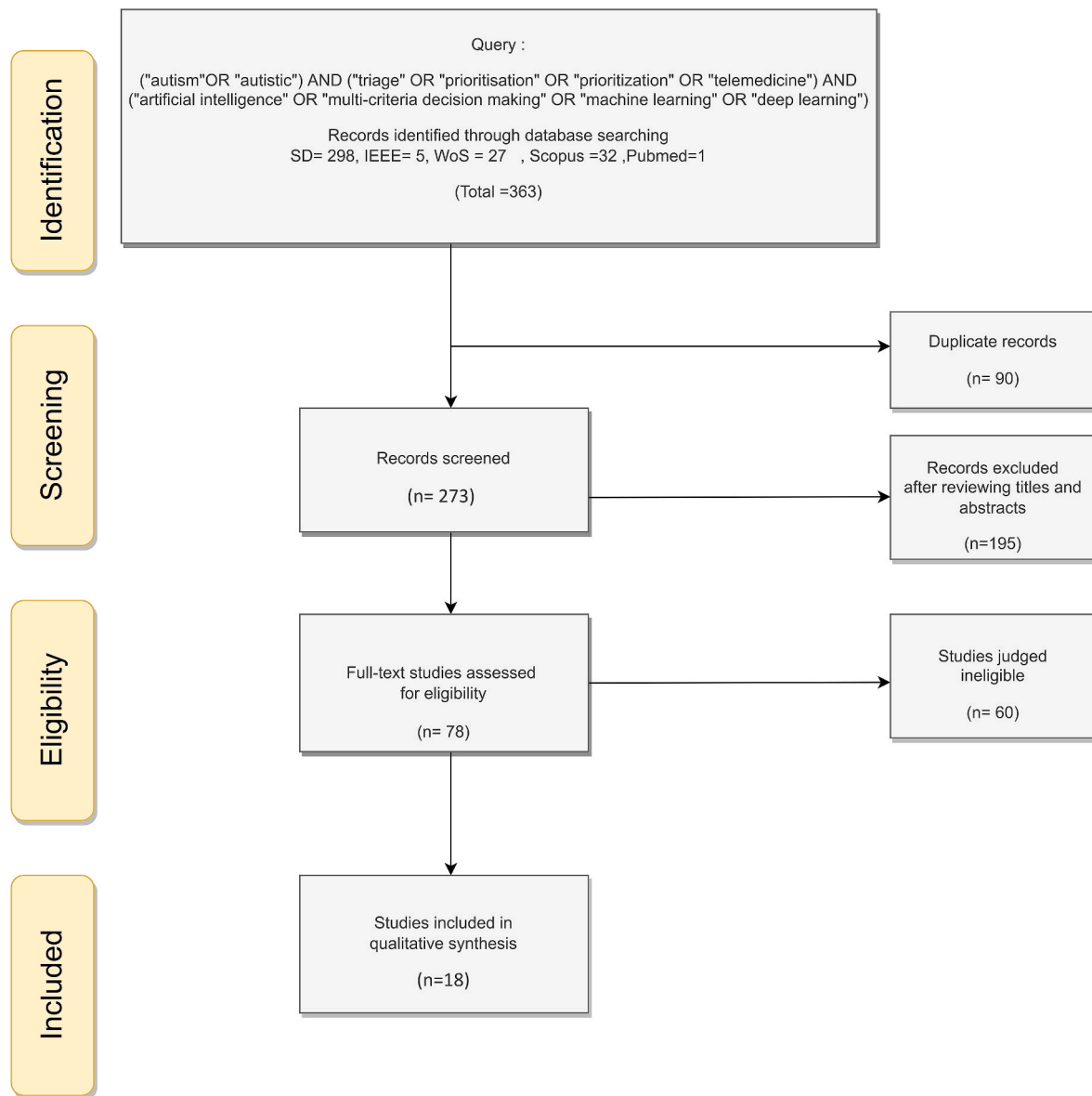


Fig. 1. Schematic of approach for identification, screening and inclusion of relevant studies.

of the published articles that were selected according to our inclusion criteria is as follows: SD (3 papers), WoS (4 papers), IEEE Xplore (1 paper), Scopus (9 papers) and PubMed (1 paper).

3.1. Annual production

Fig. 2 shows the relation between the year of publication and annual production. Most publications during 2020 were approximately 50% or (9/18) papers. Most of the papers focused on prioritising ASD risk genes. Followed by the year 2021 were 27.7% or (5/18) of publications during the year. Publication papers decreased before 2020 were focused on triage and prioritisation of ASD using AI techniques.

3.2. Country-specific production

Fig. 3 illustrates the collected articles from seven countries. The articles generally included case studies in ASD clinics, and most studies were focused on genes depending on the Simons Simplex Collection (SSC) of genes, which is known as Simons Foundation Autism Research Initiative (SFARI), to provide more understanding and improve the

diagnosis of ASD. The highest percentage of studies were from USA (50%) or (9/18 papers) followed by China (2/18 papers), Canada (2/18 papers), Turkey (2/18 papers) whilst each of the following countries were published (1/18 paper): Korea, Denmark and Macedonia in the field of Triage and prioritisation of ASD using AI technologies.

3.3. Word cloud

A word cloud points to the most essential topics that deal with a particular subject. Fig. 4 presents the important words adopted by previous studies on triage-based prioritisation for ASD. Most papers focused on the ASD patient's aspects (i.e. autism, diagnostic, prioritisation, spectrum disorders, genes, mutations and biomarkers). The keywords refer to the words that interest the authors who investigated the prioritisation of ASD. High-performance AI techniques in prediction and diagnosis have made less progress in diagnosing and prioritising autism patients based on triage. The academic literature has presented ASD diagnosis work from a specific perspective and much confusion arises from the fact that AI theory is currently integrated with ASD diagnosis concerning triage and priority.

Historical Direct Citation Network

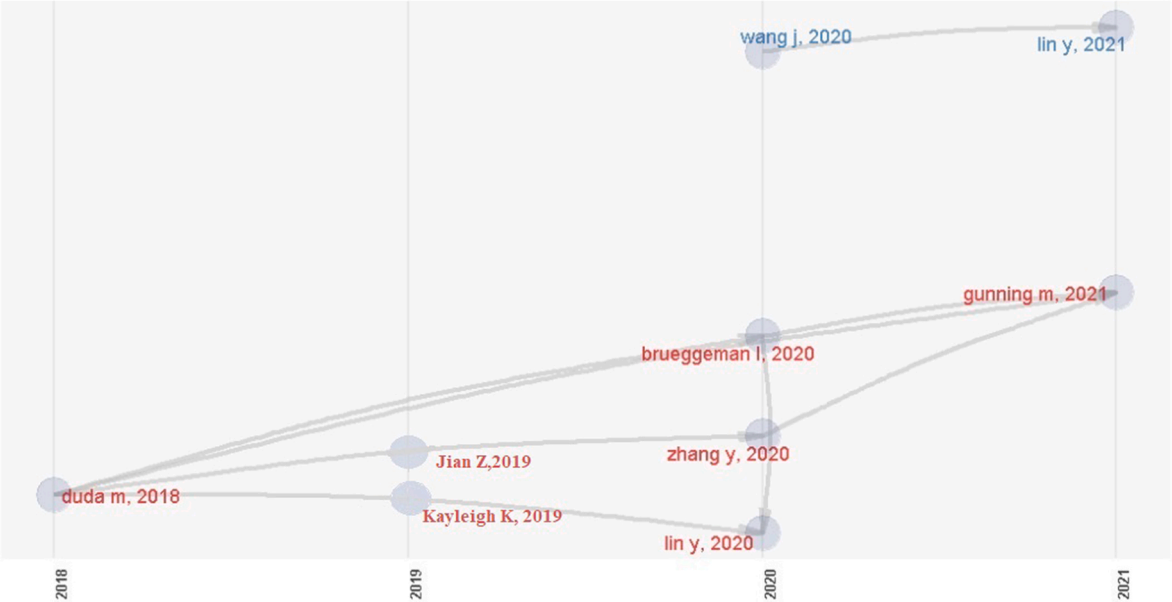


Fig. 2. Historical direct citation networks.

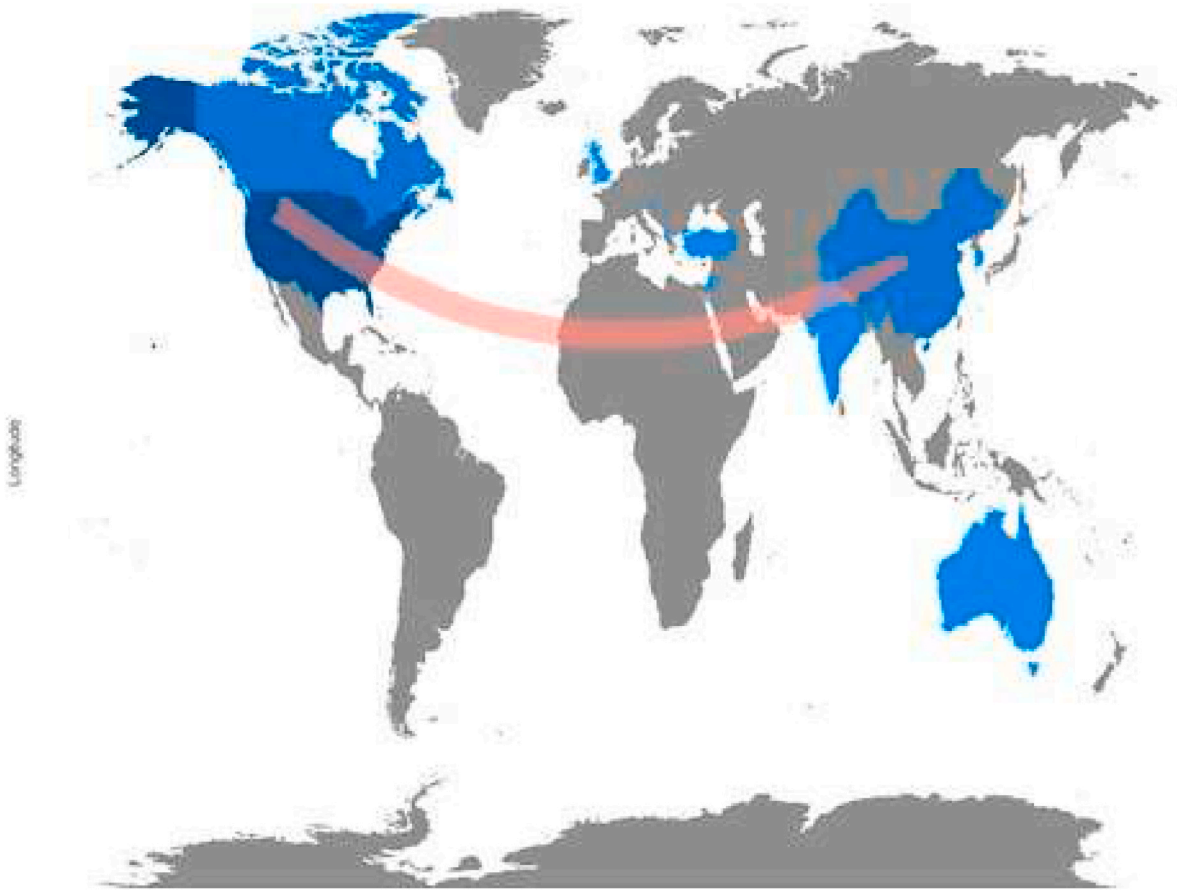


Fig. 3. Country collaboration map of Publication.

4. Search results

Schematic flowchart (Fig. 1) contains (1) Identification phase: where the query has been created to collect papers from the four reliable

databases. The mention phase collected (363) papers according to the search query and after using Mendeley application to remove the duplicated literatures the study obtains (273) articles. Then, the next Screening phase eliminated the out-of-scope articles by reading the title



Fig. 4. Word cloud.

and abstract of each of the collected articles, which results in filtering (195) articles. A full-text reading to identify the excluded articles has been achieved in the eligibility phase to obtain (78) articles. Finally, final set articles included (18) papers in the final phase (Included). The included studies that were obtained, such as prioritisation of related genes, using EEG signals to rank the related features and prioritising autism patients based on clinical triage, are close to our scope. AI technologies are used to create a more concise algorithm for ASD diagnosis and are less time consuming within these articles. In addition, the final set articles are divided into three major categories based on the aim and objective evidence across studies that fit pre-specified inclusion and exclusion criteria. The first category, which contained ($n = 3/18$) 16.66% articles, is focused on 'Triage patients based on diagnosis methods'. The second category is focused on 'Prioritisation for Risky Genes'. This category contained ($n = 12/18$) 66.6% articles and had two subcategories. The first sub-category containing (8/12 articles) was the 'Mutations Observation based', the second sub-category containing (4/12 articles) was the 'Biomarkers and toxic chemical observations'. Finally, the third category containing (3/18 article) 16.66% was the 'E-triage using telehealth'. These studies were grouped and formed into the taxonomy below (see Fig. 5).

4.1. Triage patients based on diagnosis methods

The diagnosis is the first step of any cases, and for ASD, it is essential and must be taken early for faster service. Many diagnosis methods were

used for the clinical detection of the autistic patients: ADOS, for observing the autistic patient behaviour and ADI-R with 93 question needs to assess for autism symptomatology. These types of methods spend time and cause diagnosis delay, therefore prioritising patients based on high urgency for booking to clinical diagnosis is important to accelerate the treatment. In total, ($n = 3/18$) articles among the screening papers showed triage patients based on ASD diagnosis methods according to the collected articles. In Ref. [26], this study used machine learning method to predict the subgroups of Korean ASD based on Diagnostic Statistical Manual for Mental Disorders-IV Text Revision (DSM-IV-TR) and ranks the autism ADI-R diagnostic algorithm items to prove that machine learning methods sufficiently predict the diagnosis with minimum items. They produced 10 levels of patient prioritisation based on two board-certified child psychiatrist triages where they used ADI-R, ADOS and clinical interviews. The article in Ref. [5] proposed another diagnosis method based on EEG to assist in the automatic detection of ASD. A new method for AI has been proposed using CNNs that are utilised as deep-learning algorithms to help in the automatic detection of ASD and by using a big EEG signal dataset and selecting the group of ASD patients to cover all triage levels: mild, moderate and severe. They found that signals from children with autism have a common morphological pattern, which makes them distinguishable from normal EEG. The study of [27] integrated the Rapid Interactive Screening Test for Autism in Toddlers (RITA-T) using two levels screening tools, which were used for improving the triage process for those at risk of ASD, by referring them sooner to more appropriately

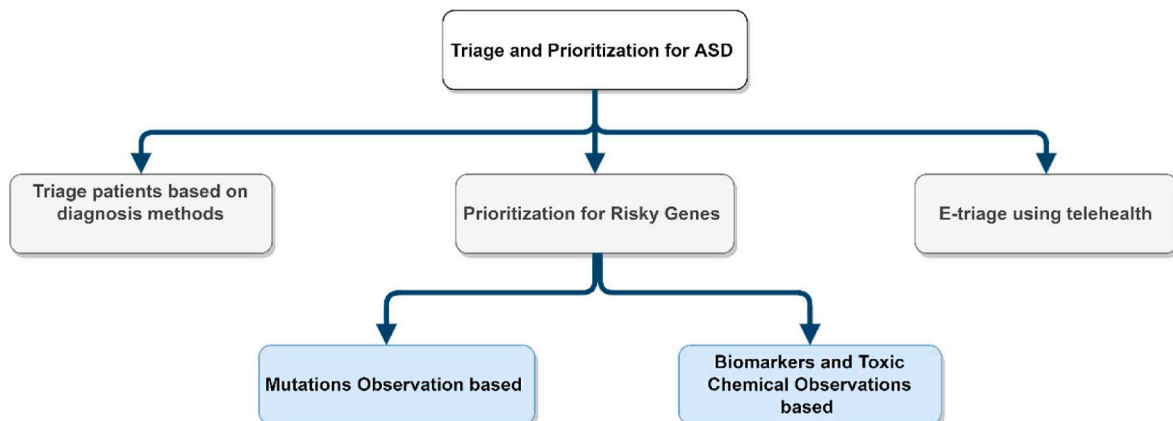


Fig. 5. Taxonomy of research literature on triage and prioritisation of diagnosis ASD based on AI.

evaluation clinics. The RITA-T detected the risky ASD patients for timely diagnosis and other health services. An unadjusted Logistic Regression (LR) model was utilised to evaluate the relationship between ASD diagnosis and RITA-T score. The RITA-T shows high correlation with autism diagnostic measures, including the ADOS-2, DSM-5 criteria for ASD and experts' opinion.

4.2. Prioritisation for risky genes

Gene mutations play an essential role in autism problems. Therefore, many studies focused on genes for autism patients to determine whether specific genes cause this kind of disorders or not. This category contains ($n = 12/18$) articles focused on prioritising genes according to their risk using some machine learning methods. This category can be classified into two subsections, the first focused on mutations that affected ASD genes, and the second concentrated on biomarkers and toxic chemical observations.

4.2.1. Mutations observation

The gene mutation is an alteration in one or more genes, and some of these variations can cause genetic disorders like ASD. This subsection contains ($n = 8/12$) publications. In Ref. [3], the study mentioned that "only a fraction of the hundreds of nominated ASD-related genes have identified de novo or transmitted loss of function (LOF) mutations that can be directly attributed to the disorder". De novo mutations impact risk for several diseases and disorders, particularly for ASD [28]. In addition, they discussed the importance of prioritisation for ASD candidate genes that help in filtering out false-positive results and for aiding the researchers to concentrate on genes that are regarded as more casualty for this disorder. By leveraging a brain-specific functional relationship network (FRN) of genes, the study constructed the machine learning model to provide a genome-wide ranking of ASD risk genes to differentiate between ASD-related and whom without. In Ref. [15], the authors also focused on identified de novo, which probably gene-disrupting (DN-LGD) mutations, they found that 353 genes nominated to autism, where 27 frequent genes likely related to autism, and 176 negative genes in siblings not affected. They proposed a bioinformatics framework and used network-based deep learning approach (PANDA) to prioritise the autism genes across the human genome, by using patterns of gene-gene interactions and topological likeness of genes in the interaction network. The study presented in Ref. [29] focused on detection problem of the risk gene by using machine learning and introducing (forecASD) approach. The authors used genome scale data as predictors to identify new genes by using similar properties to determine ASD risky genes. They found that 44 genes with more than one de novo loss-of-function mutations. In Ref. [30], the state-of-the-art machine learning approach has been developed to predict the candidate long non-coding RNAs (lncRNAs) related to ASD. The performance of the produced model indicates that RNA transcript sequence may have relevant information for ASD risk gene prediction. Besides, they used models to prioritise a list of human lncRNAs with predictable expression signals in the BrainSpan dataset. As a result, they were able to identify good candidate lncRNAs and concluded that: some de novo mutations and all familiar ASD risk genes signalling proteins may disrupt protein-coding genes and have caused ASD. In Ref. [31], the study illustrated that to foretell ASD risk genes, machine learning can be employed using features from "spatiotemporal gene expression patterns in human brain gene-level constraint metrics, and other gene variation features, and when and where ASD genes may affect during brain development". Their process of gene ranking has provided a beneficial resource for prioritising ASD candidate genes and diagnosing the patients. The study of [32] aimed to investigate whether machine learning can help in the discovery and diagnosis of disease genes. Therefore, they collected 13 published ASD gene prioritisation research and evaluated their performance using recognised and unfamiliar high-confidence ASD genes. They found that machine learning methods which do not incorporate

genetic information have restricted benefit for prioritisation of ASD danger genes. Their results elucidate that despite utilizing the complex data and developed algorithms, the Guilt by association (GBA) machine learning model for ASD had failed to outperform genetic measures of this type of disease gene. The study of [33] discovering the contribution of non-coding mutations to illness using the deep learning-based framework to predict the particulate effects and harmful impact of genetic variants. They proposed a new approach to deep learning for quantitatively defining the effect of noncoding mutations on transcriptional and post-transcriptional regulation. Their proposed framework of genomics predictive had been clarified the role of non-coding mutations in ASDs and mutations prioritisation as well. The authors in the article [19] tried to "investigate the genes and molecular mechanisms associated with the occurrence of seizures in ASD". They accomplished gene prioritisation by using five propagation algorithms: "simple random walk, heat kernel diffusion, PageRank with priors, HITS with priors, and K-step Markov".

4.2.2. Biomarkers and toxic chemical observations

The aim of autism research on genetic and chemical aetiology is: to find out the validated biomarkers, for facilitating the early autism diagnosis, prediction of healing response, pursuing progression, evaluating risk, and medical development. This subsection contains ($n = 4/12$) articles. Besides, studies in this subsection focused on how biomarkers and toxic chemicals can take place in ASD. The cause of many mental disorders like ASD is exposure to environmental toxicants (like mercury, toluene, lead, PCBs, and arsenic). In Ref. [2], they used signal processing and machine learning methods that integrated to analyse "microarray gene expression data from homogeneous groups of children diagnosed with classic autism, excluding the autism with regression and high-functioning autism's syndrome". The major focus of this paper was gene expression pre-processing, which depending on Fractional Fourier Transformation, and the acquired data are also utilised for selecting biomarkers (the most informative genes). The study of [4] proposed a new multi label classification (MLC) model to determine the autism risky genes and toxic chemicals on an extensive dataset. The article proposed a CNN-based MLC framework to prioritise the autistic risky genes and chemical toxicity that is causative and its subtypes. In Ref. [12], the study constructed predictive models to promote the ASD diagnosis and cell type specific ASD-associated genes and gene sets prioritisation that can help as biomarkers for ASD. They studied the importance of genes in the cell type-specific predictive model through prioritising of genes. In Ref. [34], the study produced a predictive solution for ASD risk genes to identify the temporospatial regions of the brain structures at different developmental time points. The study aimed to explore the specificity of ASD gene expression in the brain for helping diagnosis and detection in possible ASD in the future. They prioritised the temporospatial features based on their accuracy in prediction contribution; therefore, they identified top ten ranked temporospatial regions related in ASD aetiology.

4.3. E-triage using telehealth

The E-Triage system is essential for identifying which patients are appropriate for telemedicine or book for clinical visit and prioritising patients according to their high urgency needs. In this category, the triage of autistic patients through telemedicine techniques has been focused and discussed. In Ref. [10], the study suggested that telehealth consulting in diagnosis and early intervention systems could have a positive impact for diagnosis evaluation and the ability of caregivers and families to schedule appointments and clinical visits as well. In Ref. [9], this study aimed to provide instances about how autism clinical centres adjusted their traditional approaches to conduct their diagnosis assessment for ASD during any pandemic like Covid-19 and how shared their challenges to other centres for benefit. They suggest that all patients would be triaged based on their high urgency need for booking on two

telehealth visits; first appointment for visiting with a psychometrist and second one for visiting with a licensed psychologist. In the study of [35], a new mobile-health screening tool has been developed by Cognoa, their results suggested that this novel ASD screening Cognoa has potential psychometric characteristics that will support ASD screening and clinical triage tool. They used a machine learning algorithm on the parent questionnaire scores to produce evaluation based on questionnaire results of triage (i.e., low, medium-non ASD, medium, elevated).

5. Discussion

In this section, three important aspects of the literature behind the triage procedure and priority contexts, namely, motivations, challenges, and limitations of using AI and recommendations, are discussed to mitigate these difficulties to improve triage and prioritisation of patients.

5.1. Motivations for ASD triage and prioritisation

This section discusses the advantages of triage and prioritisation reported in the academic literature. Three main benefits can be identified from the collected articles as follows:

- The triage and prioritisation concepts can reduce cost and time consumption to provide an early diagnosis for autism patients. The conventional clinical diagnosis of autism using ADOS requires between 60 and 90 min for initial DSM-IV-TR assessment, due to variation in reliability between residents [16] and the ADI-R assessment of autistic symptoms. Each patient needs more than 2 h to complete the triage and/or priority process. For this, the authors of [26] applied “top-ranking items for ASD diagnosis in the Korean population” and can reduce the time to ASD diagnosis. They produced 10 levels of patient prioritisation based on two board-certified child psychiatrist triage where they used ADI-R, ADOS and clinical interviews. Also, the long waiting time for service, clinical appointments and higher costs are improved. Accordingly, several studies attempted to overcome these issues to reduce the cost and time consumption for diagnosis. To overcome the expensive and time consumption in the process of genetic and genomic studies were usually performed to identify ASD risk genes and numerous generated potentials nominated genes, and difficulty in validation as well. Therefore, a study [12] constructed a predictive model that could promote the diagnosis of autism and prioritised cell-type-specificity that could be used as potential biomarkers of ASD. The machine learning algorithms can identify and predict the genes involved in the disorder better than depending only on unclear traditional genetics that consume time in diagnosis [32]. Understanding the association of genes related to the disorder/disease aids better diagnosis and treatments which is better than costly experiments and time-consuming during the evaluation of the genes [15]. Triage process for children with ASD can help sooner by referring them to clinical appointments for ASD evaluation and can reduce the wait times for both types of children (who have the ASD risk and one’s non ASD risk) [10,27,35]. In conclusion to the above benefits from reducing cost and time consumption, the triage and priority can identify ASD risk and nominated genes, leverage machine learning in reducing time and cost during ASD diagnosis, and understand the association of genes related to the disorder to obtain diagnosis.
- The triage and prioritisation can improve therapeutic strategies and aid autism patients with daily activities. Determining “the brain temporospatial regions where the risk genes are expressed in ASD patients” could improve the therapeutic strategies for better healthcare and treatment for ASD risky patients [34]. Besides, releasing the association of genes for ASD patients through priority tools helps in designing therapeutic strategies and better diagnostic situations [15].

- Triage and prioritisation focusing on autism case evaluations, which are important to identify the appropriate treatment for patients and prioritise the certain patients considered ‘urgent’ than other ASD patients with less symptoms. The evaluation of the status of each ASD patient, which prompts the clinicians to consider and implement diagnostic processes innovatively to adapt the situation and serve the patients’ need, is essential during Covid-19 and any pandemics [9].

5.2. Challenges for ASD triage and prioritisation

Several concerns detailed in studies related to diagnosis, prioritising and triage patients are shown in (Fig. 6). The identification of all risky genes that contribute to ASD risk are still concerning, and no standard set of risky genes has been identified [15,29,32]. Although many studies attempted to determine them, they overlook potentially instrumental mutations with small to medium effect sizes [3]. No laboratory of neuroimaging tests or even any biochemical markers can currently be utilised as a diagnostic standard for ASD [4]. “The causal variants and the specific cell types in which the diseases-risk variants may be active are unclear” [12]. A strong clue of clinical and genetic overlap was observed between ASD and other types of neurodevelopmental disorders and represented by a spectrum of disorders, such as overlapping with epilepsy or high-functioning autism [2,31]. The age of patients that is covered has a gap due to the limited available data for ages including 31–48 months of age [35]. However, the evidence level of using EEG monitoring has not yet reached the recommended level for diagnostic or even screening of all ASD [5]. The improvement of identification of ASD earlier is still concerning for clinical, besides, the limitation of ASD clinicians, causes overwhelming, due to the volume of referrals, thereby enabling patients to get much time until evaluations [27]. The prioritisation may vary according to the availability of human equipment and resources, as well as some policies that are sensitive to domestic problems in contrast to the goal of ASD evaluations [9]. In conclusion, four main challenges were encountered: there is an ambiguity of identification of risk genes, and the overlapping with other neurodevelopmental disorders could be an open issue. The data limitations are still a valid concern, especially for the lack of medical and laboratory data, besides the issues of early diagnosis and poor supplementary services (Fig. 6).

5.3. Recommendations for ASD triage and prioritisation

This section illustrates the recommendations of all collected studies that are related to the prioritisation and triage concepts, as well as several future opportunities to evaluate the tools related to ASD, including studies for improving the accuracy of standard assessment tools.

- **Recommendations to implement the proposed models:** Some studies constructed predictive models for autistic diagnosis and others ranked their results, such as prioritised finding risky genes and recommended to replicate their finding for future work. Researchers mentioned their constructed predictive models for promoting the diagnosis of ASD and may use the “cell type-specific ASD-associated genes and gene sets” as prospective biomarkers of ASD [12]. Other study recommended replicating their findings (the Rapid Interactive Screening Test for Autism in Toddlers) by other researchers for various populations of children and families with autism [27]. Novel genetic approaches involved in neurodevelopment disorders is highly recommended to help and support autism research [34]. The whole genome ranking had been proposed as a useful resource for ASD gene discovery effort [3]. The study constructed a novelty of identifying ASD risk genes and toxic chemicals that can lead to better healthcare decisions [4]. Another study proposed as a starting point to produce new suggestions to investigate presently appreciated in the molecular aetiology of autism [29]. Hybrid deep lightweight features extracted from EEG signals are used and intended to be

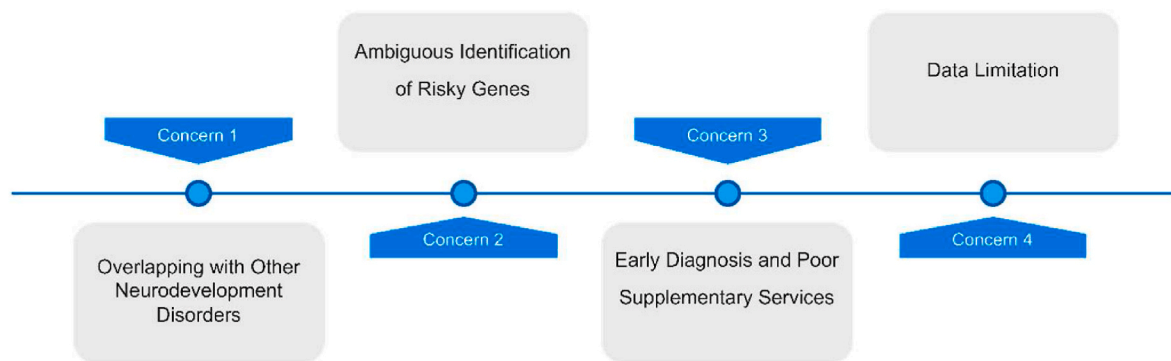


Fig. 6. Challenge categories for prioritisation and triage.

applied in the future for the early detection of ASD in clinics [5]. As a testing instrument for ASD, a proposed study (Cognoa) has potential psychometric advantages that may aid in maximizing the identification of likely ASD cases and minimizing false positives. Therefore, the resources can be allocated efficiently, and young children can receive early intervention services [35].

- **Recommendation to investigate ASD risky genes:** According to literature, no specific risky gene set finding is attained yet, but several studies had classified some genes as risky and contributed in autism. In Ref. [31], the authors recommended further investigating needs in the future to determine whether their ranked gene system can be specified only for ASD. The study of [32] focused on “developing and validating statistical models for genetic association, specifically for the association between rare variants and disease”, to discover illness genes.
- **Recommendations to utilize AI for ASD diagnosis:** The AI can provide a quick diagnosis in medical science before the final diagnostic, which depends on expert decision [26]. The use of sophisticated modelling and learning algorithms is highly recommended to remarkably describe the genetic architecture of complicated disorders and diseases [12,15].
- **Recommendations for data availability:** The data limitation can slow down the development of research in specific fields. The “high-quality gene expression profiles” must be more available publicly in the future to facilitate accurate developing models for autism risk gene prediction [30,34].

6. ASD: critical analysis and research gaps

This study conducts a comprehensive analysis of scientific mapping and discusses current state and open issues that aid in the implementation and improvement of the recommended solution of the ASD research direction. No clear identification of the contribution of AI, intelligent decision-making and machine learning is provided to evaluate and prioritise autism patients according to their health severity. However, most studies have only focused on prioritising risk genes, and no clear method has been applied to assign proper weights to gene types or even to identify influential genes in the context of intelligent decision-making methods. The integration of medical tests and demographic criteria can play a vital role in the detection of triage severity levels, and minimal attention has been paid to the prioritisation solution for the autism patients. In addition, no attempts have been made to develop a clinical triage method for ASD patients. Five complex and important open issues associated with the current topic of this study are stated in this section.

6.1. Available ASD datasets and resources

Several studies have focused on prioritising ASD-risk genes based on the online dataset (SFARI Genes) that classified many genes associated

with autism, that is, from high-confidence genes to syndrome. Other studies focused on examining ASD patients according to clinical diagnostic measures, such as ADOS and ADI-R [26,27] or by using EEG [5] to collect data (Table 2).

As shown in Table 2, the studies of [3,15,29–32] used SFARI genes as high confidence. In Ref. [14], the study used 65 reported genes from Ref. [36]. In addition, the true negative genes from non-ASD genes were created by Ref. [37], and these genes were related to non-mental diseases. In Ref. [32], the study has also used the second high-confidence ASD risky gene set compiled from three extensive Transmission and De Novo Association Analysis (TADA) studies [36,39,40]. The study of [29] added 1000 genes that are not listed in the SFARI gene database as negative examples. In Ref. [34], the dataset contained 13 developmental stages, from 8 weeks to 8 years, collected from 26 brain structures that were reported in BrainSpan atlas, the data consisted 732 samples that were risk genes of ASD = 366 and non-ASD genes = 366 on different time points of 26 brain structures and developmental stages. In conclusion, 10 out of 12 studies have been focused on gene datasets. These studies are depending on either genetic or social databases.

ASD refers to a mental disorder. It can also have social aetiologies. Thus, a sociodemographic dataset that consists of medical and social criteria that influence ASD have not been used in terms of triage and priority for autism patients. The affected criteria that can be used for ASD needs more investigation. In addition, no study uses medical and laboratory test criteria for the triage and prioritisation of patients [10, 26]. Laboratory and clinical data, such as D3, B12, Zinc, T3, T4, TSH, GSE, patient’s blood type, maternal blood type and father’s blood group, need to be considered [42,43].

6.2. Application of used traditional machine learnings and deep learning

AI science is implemented with multidisciplinary studies that aid to build more precise results. Thus, several articles in autism utilised machine learning methods (Table 3).

Study [3] trained and optimized five different ML models: SVM with linear kernel, RF, extremely randomized trees, bagging ensemble of RF and AdaBoost ensemble of RF. The training approach consisted of stratified fivefold cross-validation with a nested grid-search. In Ref. [30], different ML algorithms, including SVM, RF and logistic regression (LR), were tested. The developmental brain gene expression profiles and RNA transcript sequence compositions were used as features for model construction to predict and prioritise ASD-associated candidate lncRNA. In Study [15], the authors proposed a bioinformatics framework by using PANDA and the prioritisation of autism genes with the comparison of PANDA’s measure precision with SVM, RF and LGP being presented. Another study of [27] has used RITA-T to identify those at risk for ASD in a timely ASD diagnosis evaluation and other services. The RITA-T is highly correlated with autism diagnostic measures, including the ADOS-2, DSM-5 criteria for ASD and clinician judgment.

In conclusion, machine learning has been widely used to determine

Table 2

Available ASD datasets utilised with machine learning methods.

Ref.	Dataset description	Resource availability
[26]	First dataset = 1269 Second dataset from the first one but for children older than 2 years = 539	N/A
[31]	The confidence genes for true positives came from SFARI and 65 reported genes from Ref. [36], besides, the true-negative genes from non-ASD genes were created by Ref. [37], and these genes were related to non-mental diseases	SFARI dataset https://www.sfari.org/resource/sfari-gene/
[29]	Used SFARI high confidence genes were utilised as positive instances, all models were trained, besides, 1000 genes not found in SAFRI gene DB as negative instances.	SFARI genes https://www.sfari.org/resource/sfari-gene/
[12]	To analyse gene set, three types of 'Annotated Gene Sets from the Molecular Signatures Database (MSigDB)' [38] were utilised.	Dataset available on https://www.science.org/doi/abs/10.1126/science.aav8130
[34]	The dataset contained 13 developmental phases from eight weeks until eight years collected from 26 brain structures were reported in BrainSpan atlas, the data consisted of 732 samples were ASD risky genes = 366, genes of non-ASD = 366, on different time points of 26 brain-structures and developmental stages	Supplementary data shared online https://academic.oup.com/bioinformatics/article/32/23/3611/2525615?login=true
[32]	To evaluate their algorithm result they compiled two ASD gene-sets, using SFARI Gene 2.017 as a well-known high-confidence source database ASD genes, and another high-confidence ASD risky gene set compiled from 3 extensive Transmission and DeNovo-Association-Analysis (TADA) articles [21,39,40].	SFARI Gene 2.017 https://gene.sfari.org/data-base/gene-scoring/
[3]	To obtain the most precise predictions of genes associated with autism, we used a high set of 143 genes known to be involved in autism.	SFARI Genes https://gene.sfari.org/databases/human-gene/
[30]	The positive instances of data dataset included 604 of ASD risky genes reported as high confidence genes composed from the SFARI database categories as following: S, 1, 2 and 3. The negative instances included 1,594	SFARI Genes https://www.sfari.org/resource/sfari-gene/ And https://academic.oup.com/bioinformatics/article/32/23/3611/2525615?login=true#supplementary-data

Table 2 (continued)

Ref.	Dataset description	Resource availability
	represented a non-ASD genes collected from a study [41]	
[15]	The data had collected the set of autism genes by utilizing Databases SFARI-Gene 2.0 besides Online Mendelian Inheritance in Man (OMIM).	SFARI Genes https://www.sfari.org/resource/sfari-gene/ and OMIM https://doi.org/10.1093/nar/gki033
[4]	Using different databases. They identified 2 types of ASD high risk status which is chemicals as well as genes.	Supplementary data available
[27]	N/A	Along of 12 months, 81 toddlers were enrolled in this project. Whereas 57 diagnosed as ASD and 24 had a diagnosis of non-ASD/DD
[33]	The data had contained 1,790 ASD simplex families. Where obtained from (SFARI)	SFARI Genes

the accuracy metrics used in diagnosis. The ability of ML to extract information from the large collections of data is promising for ASD research and their future endeavours in diagnosis and triage-based prioritisation of ASD patients [19]. According to the literature, several ML methods, such as SVM, RF and others, have been used in diagnosing ASD or distinguishing the risk genes of ASD and non-ASD genes. SVM method has been frequently used in literature and gains a good accuracy amongst all other methods for predicting autism cases. The reason is that SVM uses a technique called a kernel trick to transform ASD data. Based on these transformations, an optimal boundary was observed amongst the possible outputs. Thus, different types of kernel functions have been used and produced results with high accuracy. In addition, using SVM method during ASD studies, it is used and proven as an accurate method in multiple aspects, such as its use during "enhanced normalization and evaluation of time-course and multi-class metabolomic data" [51,52] and as a SVM classifier to ensure the appropriate detection of gene signature and showed 'a good performance in classifying microarray data' [53]. On the contrary, minimal attention has been produced on the basis of other AI techniques, such as intelligent decision making, particularly multi-criteria decision making, to provide proper weights for the gene contributions or for developing triage method for ASD patients. In other fields, such as heart chronic disease, the literature developed various triage using decision-making techniques. However, for the ASD field, no method has been developed for this regard yet. Besides, the prioritisation of autism patients to detect the most severe case and ranking other patients is required. In these contexts, the telemedicine application can be adaptive for monitoring and providing the patients with suitable healthcare services from medical centres or hospitals. Deep learning is a complex part of ML, and it uses a complicated structure of algorithms modelled on the human brain. Therefore, it enables the unstructured data to be processed like images. In the medical field, few literatures covered the objective of prioritisation of ASD and triage with deep learning approaches DL according to our knowledge, there are few literatures covered the objective of prioritisation of ASD and triage with deep learning approaches. While many of studies were focused on diagnosing and distinguishing the ASD from healthy controls, most of studies used NN deep learning models in their study [44–50,54]. Whilst several deep learning algorithms were utilised in several medical aspects. It can interpret images, such as X-ray, MRI scan for accurate diagnosis of the disease, such as the proposed deep learning algorithm with a protein-encoding strategy to control the false detection average in protein-function-annotation [55] or can uses to detect the cancer

Table 3
Machine learning methods and metric results in the literature.

Ref.	ML and DL techniques and results	Matrices and notes
[26]	Multiclass decision forest (multiple decision trees)(Bagging)	1- 1st set, tree = 4 Overall accuracy 94% 2- 2nd set, tree = 4, 5 attribute Overall accuracy 98% AUC-ROC for each fold area under precision-recall curve (PRC)
[31]	AUC-ROC 1- LR 84% 2- SVM 80% with Gaussian kernel 3- RF 85% 4- Gradient boosted trees 86% PRC 1- LR 48% 2- SVM 34% with Gaussian kernel 3- RF 50% 4- Gradient boosted trees 55%	For sensitivity and specificity respectively: 1- LinearSVM = 95%, 70% 2-KNN = 90%, 84% 3- RF 98%, 92% 4-Naive Bayes 100%, 99%
[12]	Partial least squares regression Principal component analysis and multiple regression	NA
[34]	SVM-based risk gene prediction method ASD-Risk	By using 10-fold cross-validation, the ASD-Risk achieved accuracy = 81.83%, sensitivity = 84%, specificity = 79%, AUROC = 84%
[2]	1- Linear SVM and nu-support vector machine (nu-SVM) 2-KNN 3- RF 4-Naive Bayes 5-Fractional Fourier Transformation (FRFT) using MATLAB 6-Entropy - based method	5- NuSVM = 90%,90% Studies in this paper do not use information from ASD genetic association studies, but They use machine learning algorithms to distinguish ASD from non-ASD risk genes using other types of non-genetics data. ROC-AUC for all classifiers over five-fold cross validation 1- SVM = 83% 2- RF = 86% 3- extremely randomized tree = 86% 4- bagging ensemble = 87% 5- Adaboost = 82%
[32]	NA	They applied several metrics for each ML algorithms: (accuracy, sensitivity, specificity, ROC AUC, PR AUC and MCC) On expression features, Expression sequence features The accuracy for each method: 1- PANDA = 89% 2- SVM = 61% 3- RF = 67% 4- LGP = 63%
[3]	1- SVM 2- RF 3- Extremely randomized tree 4- Bagging ensemble 5- Adaboost	The MLCNN model earns the highest F1-scores of 86% and 80% for risk genes in ASD and AD, respectively. NA
[30]	LR, SVM, and RF	96.44% was obtained with the cubic SVM classifier
[15]	PANDA (GNN graph neural network), SVM, RF, LGP	NA
[4]	Using a multi label convolutional neural network (MLCNN), and implement five-fold cross validations (CVs)	NA
[29]	RF	NA
[5]	Convolutional neural networks (CNNs) were used as deep learning algorithms	NA
[27]	Unadjusted LR model	NA
[33]	Deep convolutional-network-based	NA
[44]	3D-convolutional-neural-network (CNN)	Using overlapped maps: accuracy = 70.5%, Sensitivity = 74%, Specificity = 67% And Using ICNs: accuracy = 45%, Sensitivity = 51%, Specificity = 39%
[45]	Deep neural network (DNN) with the advanced features and (DNN) with auto encoder (AE)	(DNN) with the advanced features = accuracy of 76.2% and the

Table 3 (continued)

Ref.	ML and DL techniques and results	Matrices and notes
		receiving operating characteristic curve (AUC) of 79.7%, and (DNN) with (AE) = accuracy of 79.2% and the AUC of 82.4%
[46]	Convolutional-neural-network-based diagnostic classifier	The ROC-AUC of the model = 95.5%, an accuracy = 88%
[47]	CNN with a brain imaging dataset	accuracy = 70.2%
[48]	Applied lasso, group lasso, and a new technique, Sparsely Grouped Input Variables for Neural Network (SGIN)	NA
[49]	Deep Neural Networks (DNN)	mean accuracy = 50.74%
[50]	New deep learning approach had modelled which included multiple neural network NNs called: random NN cluster	1-Probabilistic-neural-network (PNN) accuracy = 90% 2-(DNN) accuracy = 86.36% 3- deep-learning to diagnose ASD accuracy = 70% 4- adopted-deep-learning to classify the ASD patients and TC accuracy = 70%

extensively. Besides using it in non-medical aspects, like utilizing CNN to annotate of bacterial type IV secretion system [56]. While deep learning needs a large amount of data. Furthermore, the accurately models and more powerful will be needs more parameters, where it requires more data and cannot handle multitasking. On the contrary, little attention has been produced on the basis of other AI techniques, such as intelligent decision making, multi criteria decision making, for giving proper weights for the genes contributions or for developing triage method for ASD patients. In other fields such as heart chronic disease, the literature developed various triage using decision making techniques. However, for the ASD field, no method has been developed for this regard. Besides, the prioritisation of autism patients to detect the most severe case and ranking other patients are required. In these contexts, the telemedicine application can be adaptive for monitoring patients and providing the suitable healthcare services from medical centres or hospitals.

6.3. Gene contributions towards machine learning

Gene is a basic functional and physical unit of inheritance where deoxyribonucleic acid (DNA) was represented as a molecular repository of genes that encodes the structure of a gene product, either RNA or protein [57]. Meanwhile, a gene mutation corresponds to a modification in one or more genes. Some mutations can lead to genetic disorders or illnesses [58]. Therefore, many scholars researched the effectiveness of genes and mutations on ASD. The ASD is extremely heritable with the association of several genetic syndromes, such as 'Fragile X and Rett syndrome' [4]. Accordingly, several studies found that the de novo or transmitted loss of function (LOF) mutations could disrupt protein-coding genes and were regarded as ASD aetiology [30]. Several studies leverage machine learning approaches for prioritising high-confidence gene candidates in ASD. The enhanced implementation of effective interventions to predict the highest confidence gene sets that affect ASD [12]. Accordingly, AI tools and methods play a vital role in revealing many risky genes that may contribute to ASD [59]. According to the literature, several ML methods had been used to distinguish the risk genes of ASD (Table 4).

As shown in Table 4, each of the mentioned studies had produced a top-ranked ASD gene set by using known ML approaches to identify and prioritise the genes of ASD. All listed studies above had attempted to label genes into risk and non-risk genes by using several ML methods, such as SVM, RF and LR. According to Table 4, two genes have been replicated in studies as high-confidence ASD genes: MYCBP2 and PTEN gene. In conclusion, different studies had proposed various ML prediction models to produce ASD genes whilst specific gene sets have not been

Table 4
ASD candidate genes according to the literature.

Ref.	Genes and mutations involved in ASD	ML algorithms
[31]	"NBEA, HERC1, TCF20, ZYG11B, HECTD1, CAND1, and MYCBP2"	LR, SVM with Gaussian kernel, RF and Gradient boosted trees.
[12]	"We found that AST-PP, IN-PV, L2/3, L4, and L5/6-CC may be more vulnerable in ASD, the possible association between BCYRN1 and ASD"	Multiple regression and principal component analysis (PCA)
[30]	"Top five candidate lncRNAs highly prioritised to be associated with ASD (XIST, AC132217.1, LINC02607, AC011603.4, AC020907.2)"	SVM, RF and LR
[15]	"Top-ranked unlabeled genes using PANDA (RUNX1T1, MAGI2, GRIA3, MYCBP2, AKAP6, PTPRD, AUTS2, MYO9A, ABI2, PLXNA2)"	GNN, SVM, RF, LGP
[4]	Researchers found that at least 1% of people with ASD and macrocephaly have mutations in the PTEN gene.	MLCNN
[3]	Cluster 1 and 2 represented the important biological procedures were contributing to ASD, cluster 3 had been included the ASD genes like "BRAF, PTEN, and NTRK1 which involved in synaptic transmission and learning/memory", while cluster 4 had "ASD genes CHD7, CHD8, and CTCF"	SVM, RF, extremely randomized tree, Bagging ensemble and Adaboost.

clearly addressed yet [19].

6.4. Gap analysis directions

This study aims to show the importance of prioritisation based on specific triage for patients who suffer from ASD and how AI can aid in this field. The literature that are collected on the basis our taxonomy can offer certain advantages. Firstly, most of the collected studies focused on reducing cost and time consumption, which are important for early diagnosis to help families from waiting a long time to be served [10,12,15,26,27,32,35]. Meanwhile, another study focused on improving therapeutic strategy, which is essential for enhancing the healthcare of patients [34,60,61]. The evaluation of autism patients is important for their prioritisation according to the urgency of their needs where triaged based on specialists decisions [9]. The study also showed some concern according to literature. The diagnosis of autism remains challenging itself due to highly overlapping ASD with other neurodevelopmental disorders, such as high-functioning autism and epilepsy [2,31]. Another concern is the limitation of data and resources because of the difficulty of diagnosing autism in the early time and the time consumption of diagnosis during clinical appointments; in addition, many electronic tests are not recommended due to their risks, thereby making the information for ASD limited and difficult to collect [5,35]. According to literature, genes and genetic mutations are considered causes of autism disorder. Studies differed in identifying and classifying dangerous genes; therefore, finding the identification of genes that cause autism is still challenging [3,4,12,15,29,32]. In conclusion to the above discussion, despite the lack of follow-up or corroboration in the literature, these findings continue to be an open issue in subsequent literature and have shown the lack of investigation triage and/or priority methods based on AI for supporting the diagnosis of ASD. To examine new diagnosis technologies and provide researchers with a clear vision of the triage and priority tools, we must first be aware of the approaches and existing limitations applied, as well as the factors and its weights, available dataset types and the recommended machine learning methods in this line of ASD. If this requirement is considered, then the construction of the diagnosis model is expected to enable physicians and medical staff to eventually analyse serial measurements in global ASD medical research. The research was analysed from three perspectives (e.g., triage of patients, priority patients and genetic contributions) based on whether the

study followed any of these trends (Table 5).

As a conclusion from Table 5, a huge gap is found in triage based on genes. The collected articles have not focused on the triage concept. Meanwhile, most of the studies were concentrated on prioritising genes to identify the ASD gene set. In Table 5, the prioritisation based on patients approached together with triage based on patients due to the related context between triage and prioritisation and the important benefits of two principles for patients' diagnosis and treatment. According to the literature analysis and taxonomy, previous studies did not cover the roles of MCDM methods in ASD. Nevertheless, MCDM techniques can be leveraged in autism disorder to determine the weight of ASD criteria and find the important features for this kind of disorder or rank the alternative of patients or genes that can be affected on ASD. Therefore, this article suggests future research that leverages MCDM techniques for ASD, which will be explained in the next section.

6.5. Research proposal on potential future direction

According to the systematic literature review analysis regarding the triage of autism patients (Section 4.1), no exclusive study has developed MCDM-based approaches that used hybrid solution that can provide triage or prioritisation for autism patients according to their severity levels. In addition, in Section 6.1, no dataset for serving this purpose is available. Similarly, a comprehensive process is generally lacking to recommend the suitable methods for the proposed hybrid approaches. Thus, the presented study describes the potential future direction of the triage and prioritise methodological aspects for ASD patients based on 'medical tests and demographic criteria' by using integrated MCDM methods.

The hybridised triaging and prioritisation of ASD patients based on MCDM solution can be achieved by detecting the health conditions of autism patients by categorizing them on the basis of the autism severity levels, which represent the triage levels. Subsequently, the ASD patients (ranking) were prioritised within each triage level. Such hybridisation is a challenging task and is not reported in literature. The evolution of this new methodology version should be dependent on a new verified ASD dataset, including the 'medical tests and demographic criteria'. This research direction presents an intelligent triage and prioritisation framework for diagnosing patients with ASD and can support remote triage and prioritisation through telemedicine. The conceptual methodology for the proposed work can be shown in Fig. 7.

Fig. 7 shows that three MCDM methods, namely, fuzzy Delphi method (FDM), which is obtained from Study [62]; IT2TR-FWZIC method, which was obtained from study [63]; and FDOSM [64], have been used in this research proposal over all the connections of these methods were combined to propose our innovation that will benefit ASD patients. The justification behind the use of these methods can be discussed as follows.

For the FDM, this method is derived from the traditional Delphi method combined with fuzzy theory to address the issues of criteria determination and standardisation [62]. According to the number and types of the 'medical tests and demographic criteria', the selection of the affected set of criteria should be achieved on the basis of specialised ASD experts. As such, fuzzy Delphi method (FDM) allows experts to provide their views consistently [65]. FDM operates by fuzzy statistics or by continuous mathematically explicit membership functions to steady the iterative process. This method is useful in unifying ASD criteria and establishing importance levels so they can be identified best when assessing medical tests and demographic criteria.

In general, for the FWZIC, various MCDM techniques were presented for the evaluation and weighting criteria in different fields, including analytical hierarchy process (AHP) [66–68], weighted product model (WPM) [69], hierarchical adaptive weighting (HAW) [70], best-worst method (BWM) [71], multiplicative exponential weighting (MEW) [72], weighted sum model (WSM) [73], simple additive weighting (SAW) [74] and analytical network process (ANP) [75]. The superior

Table 5
Directions of studies in the context of ASD.

Ref.	Triage based on patients	Priority based on patients	Triage based on genes	Prioritisation based on genes
[35]	✓			
[9]	✓	✓		
[10]	✓			
[31]				✓
[12]				✓
[2]				✓
[32]				✓
[3]				✓
[30]				✓
[15]				✓
[27]	✓	✓		
[26]	✓	✓		
[4]				✓
[29]				✓
[5]	✓			✓
[34]				✓
[33]				✓
[19]				✓
Total percentage	33.3%	16.6%	0%	66.6%

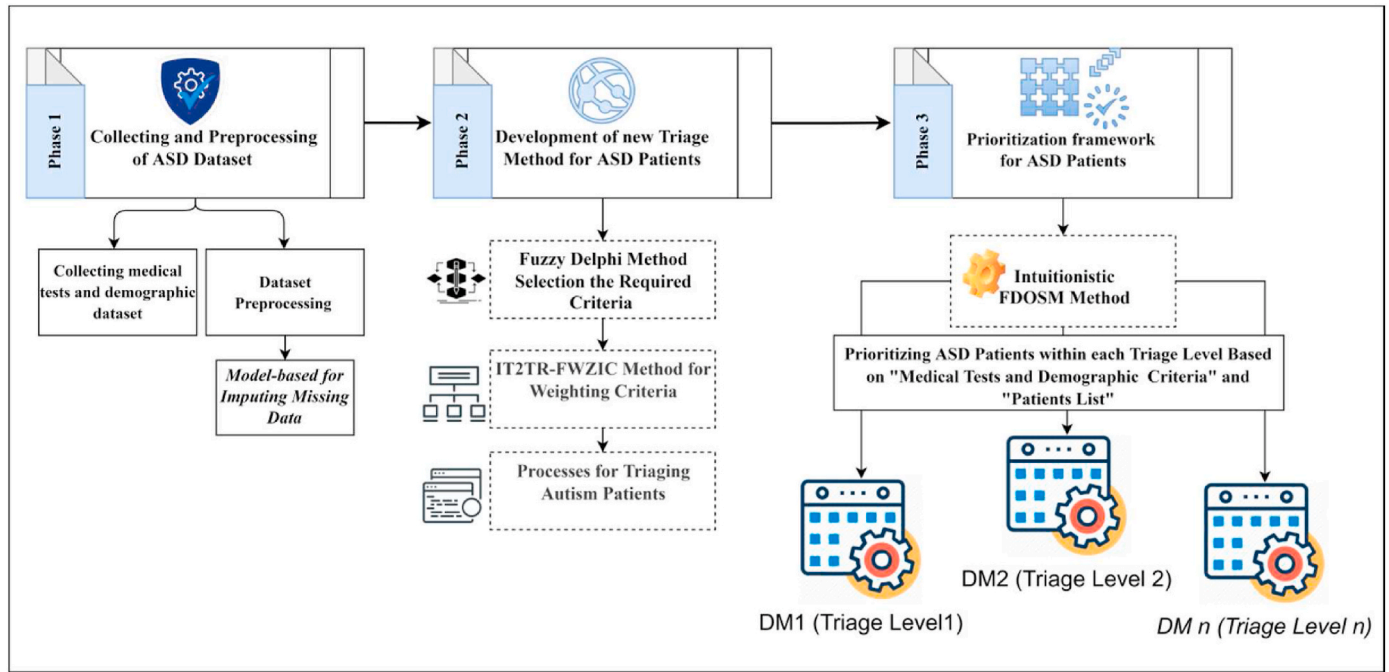


Fig. 7. Conceptual methodology phases for ASD Triage based prioritisation.

AHP, ANP and BWM methods suffer from inconsistency issues that will dramatically increase with the number of criteria in a pairwise comparison. Thus, the use of one of these methods to evaluate the selected medical tests and demographic criteria obtained from FDM is considered impractical due to these disadvantages. Many comparisons complicate the application of the model, especially in the cases of a larger number of criteria. According to some authors [76], performing completely consistent pairwise comparisons in the AHP method is almost impossible if more than nine criteria exist. This problem is usually surpassed by dividing the criteria into sub criteria, which further complicates the model. This inconsistency problem can be solved by the relatively new fuzzy-weighted zero-inconsistency (FWZIC) method [77], which determines the weight coefficients of the criteria with zero inconsistency issue regardless of the number of criteria. This method depends on the differences in the expert preferences per criterion to compute its significance level in the decision-making process. FWZIC method has been presented in different versions [77–79]. The variations of the three

versions are related to vagueness's issues. However, the latest version is [80], and this study is going to use it.

For the FDOSM, the latest and powerful method called fuzzy decision by opinion score method (FDOSM) based on triangular fuzzy number (TFN), which was proposed to consider the idea of an ideal solution, avoid two preferences, reduce the number of comparisons, define fair and implicit understandable comparisons, prevent inconsistency, reduce vagueness and yield a minimum number of mathematical operations [81–83]. FDOSM applies the idea of ideal solution and opinion matrix under a fuzzy environment with fewer mathematical equations. FDOSM provides logical decisions because it depends on the opinion of the decision matrix (the expert). FDOSM can mostly overcome inconsistency, which is the main problem in the human approach, and shortens the time consumed when performing comparisons. Therefore, this method saves data and provides a logical decision. Moreover, normalization and the weight in mathematical approach problems are surpassed. The vagueness in the data can also be solved by using fuzzy numbers.

7. Conclusion

The implication of this study will offer relevant information to practitioners, the healthcare sector and physicians about the works involved in the diagnosis of ASD and the impact of AI technique in developing and providing accurate triage and priority tools. This systematic review supports the use of machine learning to suggest predictive models that can be involved during autism diagnosis for this triple track of interdisciplinary research, namely, ASD diagnosis and gene contributions, triage/priority tools and strategies and AI theory solutions. The study found that few literatures attempted to find a method used to prioritise autistic patients based on clinical triage. According to the literature, most of the included articles have discussed the issue of prioritising autism by studying the affected genes and biomarkers, and only few have focused on studying the traditional clinical diagnostic methods and attempted to interfere with machine learning approaches to accelerate diagnostic patients. Scattered efforts on identifying risk genes for ASD disease through AI techniques have been exerted in the literature. Additionally, the ASD triage is still dependent on clinical questionnaires without any technological consideration to leverage the speed and concise tools that can be provided by AI and ML. A surprising open issue is that the early diagnosis of young patients (i.e., less than 18 months of age) with autism remains inconclusive. In addition, telemedicine applications have a low opportunity to extend outstanding solutions for remote monitoring, detection and diagnosis of autism patients of all ages. In terms of triage and priority, this study supports the application of AI and machine learning to propose a real-time predictive model that could be useful for diagnosing autism via telemedicine applications. Certainly, intelligent triage and prioritisation system are essential to detect the severity levels of patients' and identify suitable patients for telehealth diagnostic applications. In this paper, focus has been dedicated to the concept of prioritisation and how it can integrate autism patients. The prioritisation of patients based on medical triage can be an important method to be applied in telemedicine to obtain a quick and specific diagnosis and therapy if possible. Prioritising patients based on their urgent needs, is remarkable for early diagnosis and treatment. However, no decision-making method can be leveraged to prioritise ASD patients based on triage yet. Therefore, this study proposed the use of decision-making techniques for ASD patients to leverage from the advantages and implications of MCDM methods (e.g., Delphi method, IT2TR-FWZIC method and FDOSM) to determine the important features of ASD for triaging and prioritising patients according to their severity levels.

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Declaration of interest and authorship conformation

- 1) All authors have participated in (a) conception and design, or analysis and interpretation of the data; (b) drafting the article or revising it critically for important intellectual content; and (c) approval of the final version.
- 2) This manuscript has not been submitted to, nor is under review at, another journal or other publishing venue.
- 3) The authors have no affiliation with any organization with a direct or indirect financial interest in the subject matter discussed in the manuscript

Declaration of competing interest

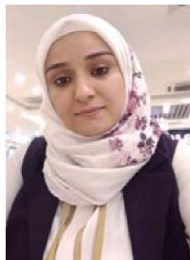
'None Declared'.

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